§3.6.

P1. Solution:

(a) A - an 7×9 matrix, $r(A)=5$.

$\dim(C(A^T)) = 5$

$\dim(C(A)) = 5$

$\dim(N(A)) = 9-5=4$ $\rightarrow$ sum $=16$

$\dim(N(A^T)) = 7-5=2$

(b) A - an 3×4 matrix, $r(A)=3$

$\dim(C(A)) = 3$

$\dim(N(A^T)) = 3-3=0$

P4. Solution:

(a) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -3 & 0 \\ 1 & -3 & 0 \\ 3 & -9 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 3 & 1 \\ -1 & -\frac{1}{3} \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$

P6. Solution:

a) $A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[P_{23}]{P_{13}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$

pivots free variables

$r(A) = 2$

i) $\dim(C(A^T)) = 2$

Basis: $\begin{bmatrix} 0 \\ 3 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

ii) $\dim(C(A)) = 2$

Basis: $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$

iii) $\dim(N(A)) = 4-2=2$

Basis: $\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ $\rightarrow$ special solutions

iv) $\dim(N(A^T)) = 3-2=1$.

solve $A^T y = 0$

$y_1\begin{bmatrix} 0 \\ 3 \\ 3 \\ 3 \end{bmatrix} + y_2\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + y_3\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = 0$

$\Rightarrow y_1 = y_3 = 0$

Basis: $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

2. $B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = R$, $r(B)=1$.

i) $\dim(C(B^T)) = 1$

Basis: $[1]$

ii) $\dim(C(B)) = 1$

Basis: $\begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$

iii) $\dim(N(A)) = 1-1=0$

no Basis

iv) $\dim(N(A^T)) = 3-1=2$

solve $B^T y = 0$

$y_1 + 4y_2 + 5y_3 = 0$

$\Rightarrow s_1 = \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, s_2 = \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}$

Basis: $\begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}$.

P.9. Solution:

(a) $[A]$ and $\begin{bmatrix} A \\ A \end{bmatrix}$

The row space and nullspace are the same.

(b) $\begin{bmatrix} A \\ A \end{bmatrix}$ and $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$

The column space and left nullspace are the same.

P12. Solution:

$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

This can't be a basis for the row space and nullspace at the same time, since the row space is always perpendicular to the nullspace.

P26. solution: B B A.

§4.1.

P3. Solution:

(a) $\begin{bmatrix} 1 & 2 & -3 \\ 2 & -3 & 1 \\ -3 & 5 & -2 \end{bmatrix}$

(b) impossible.

$2-3+5 \neq 0$

(c) impossible.

$$\left.\begin{array}{l} Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ A^T\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow [1\ 0\ 0]A = [0\ 0\ 0] \end{array}\right\}$$

$\Rightarrow [1\ 0\ 0]Ax = [0\ 0\ 0]\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$[1\ 0\ 0]\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$

$1 = 0 \rightarrow$ Contradiction

(d) impossible.

$A^TA = 0 \Rightarrow A = 0$

(e) impossible.

$(a_{ij}) = A = [a_1\ a_2 \cdots a_n] = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^m \end{bmatrix}$

$$\left.\begin{array}{l} \sum_{j=1}^{n} a_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} = 0 \\ \sum_{j=1}^{m} a^j = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Rightarrow \sum_{j=1}^{m}\sum_{i=1}^{n} a_{ij} = n \end{array}\right\rangle \text{Contradiction}$$

P6. Solution:

$A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \overset{=b}{\begin{bmatrix} 5 \\ 5 \\ 9 \end{bmatrix}}$ where $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}$

$$\left\{\begin{array}{l} y_1\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + y_2\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} + y_3\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = 0 \Rightarrow A^Ty = 0 \\ 5y_1 + 5y_2 + 9y_3 = 1 \Rightarrow y^Tb = 1 \end{array}\right.$$

$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 2 & 3 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

$\rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$

$Ry = 0 \Rightarrow s_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \Rightarrow y = c\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -c \\ -c \\ c \end{bmatrix}$

$5(-c) + 5(-c) + 9c = 1$

$\Rightarrow -c = 1 \Rightarrow c = 1$

$\Rightarrow y = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$y \in N(A^T)$!

P7. Solution:

$$\left.\begin{array}{l} x_1 - x_2 = 1 \\ x_2 - x_3 = 1 \\ x_1 - x_3 = 1 \end{array}\right. \Rightarrow \begin{array}{l} \frac{1}{3}(x_1-x_2) + \frac{1}{3}(x_2-x_3) \\ + \frac{1}{3}(x_1-x_3) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} \\ \Rightarrow 0 = 1. \end{array}$$

P9. Solution: <u>Column space</u>

<u>orthogonal to the nullspace of A^T</u>

P12. Solution:

$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow r(A) = 1$

$\dim(C(A^T)) = 1$

Basis: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\dim(N(A)) = 2 - 1 = 1$

Basis: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

These two vectors $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^2$.

$x = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = a\underset{\leftarrow x_r}{\begin{bmatrix} 1 \\ -1 \end{bmatrix}} + b\underset{\leftarrow x_n}{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$

$\Rightarrow a = 1, b = 1.$

$\Rightarrow x = x_r + x_n$ where $x_r = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, x_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

P16. Proof:

$y \in N(A^T)$ means $A^Ty = 0$

$\Rightarrow y^T(Ax) = y^TAx = (x^T\underbrace{A^Ty}_{0})^T$

$= 0$

$\Rightarrow y \perp Ax.$

P18. Solution: $\underline{A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 2 \end{bmatrix}}$.